# Appendix A: Summary of Experimental Conditions

Table 1 describes the 47 experimental conditions included in the model comparison, including the number of trials, the number of words presented per trial (Words/Trial), the number of referents presented per trial (Objects/Trial), the number of to-be-learned word-referent pairs (Items), people's overall mean accuracy ($p(o|w)$ across all intended $w-o$ mappings) in each condition (Accuracy), the standard deviation of performance (SD), and the number of participants per condition (N).

# Appendix B: Clustering experiments and models by misfit

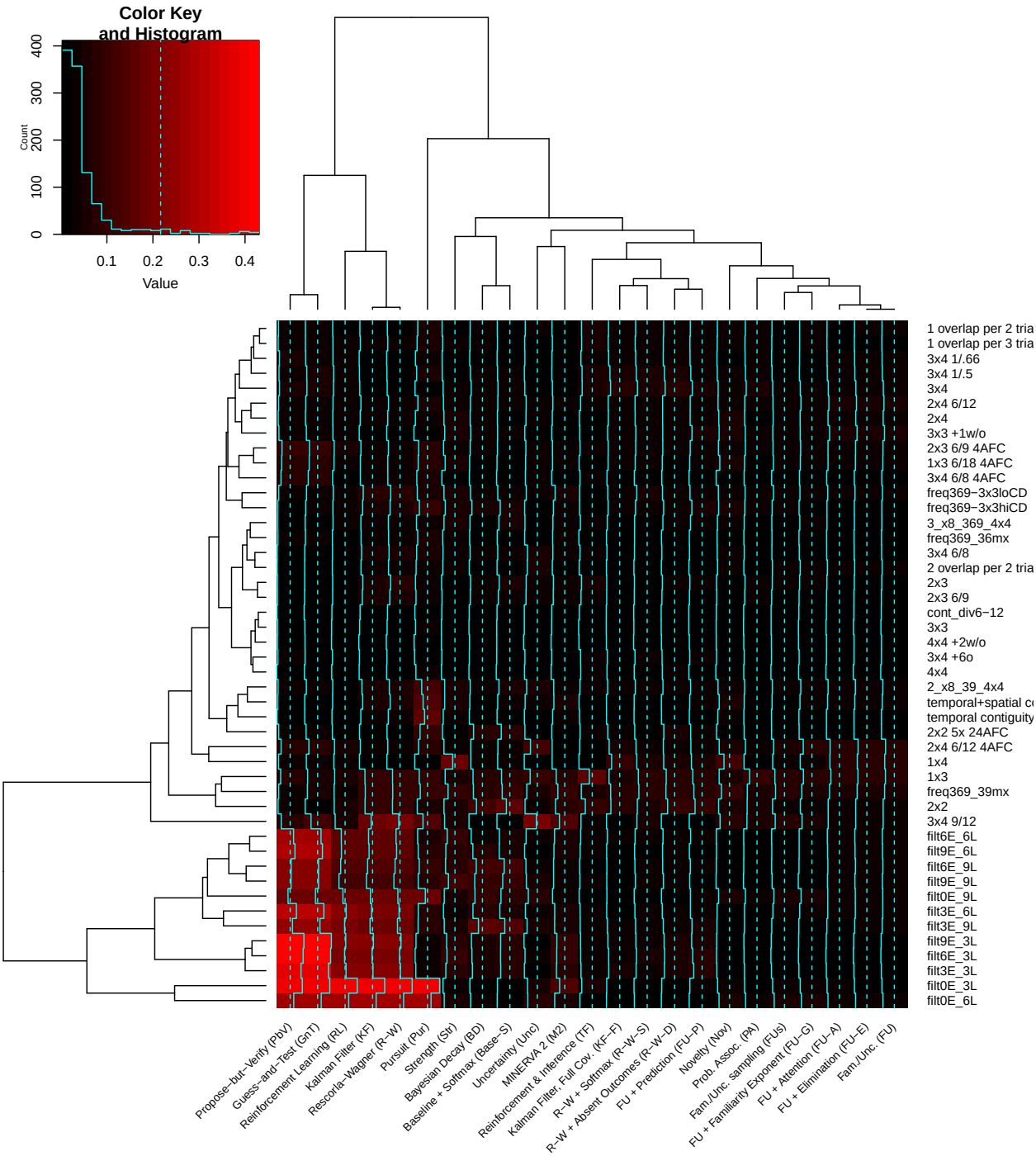
The heatmap below shows cross-validated model fit (SSE; sum of squared error) for each experimental condition, across all 23 fitted models and the 46 conditions with cross-validated fits. Note that each model has difficulty fitting at least one or more experimental conditions–and these difficult conditions vary somewhat by model. For example, the strength-biased model has particular difficulty with the `filt` conditions[1] (Kachergis et al., 2012), which present a group of word-referent pairs early in training which then systematically co-occur with particular novel late-stage word-referent pairs, testing how strictly learners will maintain a mutual exclusivity (ME) constraint. The Propose-but-Verify model also shows greater misfit in most of these conditions (except for the `filtXE_3L` conditions, which have only 3 repetitions of the late-stage pairings, and thus do not overwhelm learners' ME bias).

On the other hand, many experimental conditions are nearly equally well fit by all models, especially those that have a fixed number of repetitions per word-referent pair (e.g., 3x3 and 4x4, although 2x2 presents difficulties for some models).

[1] Except for the `filtOE_` conditions, which consist only of the late-stage pairs, with no early stage.

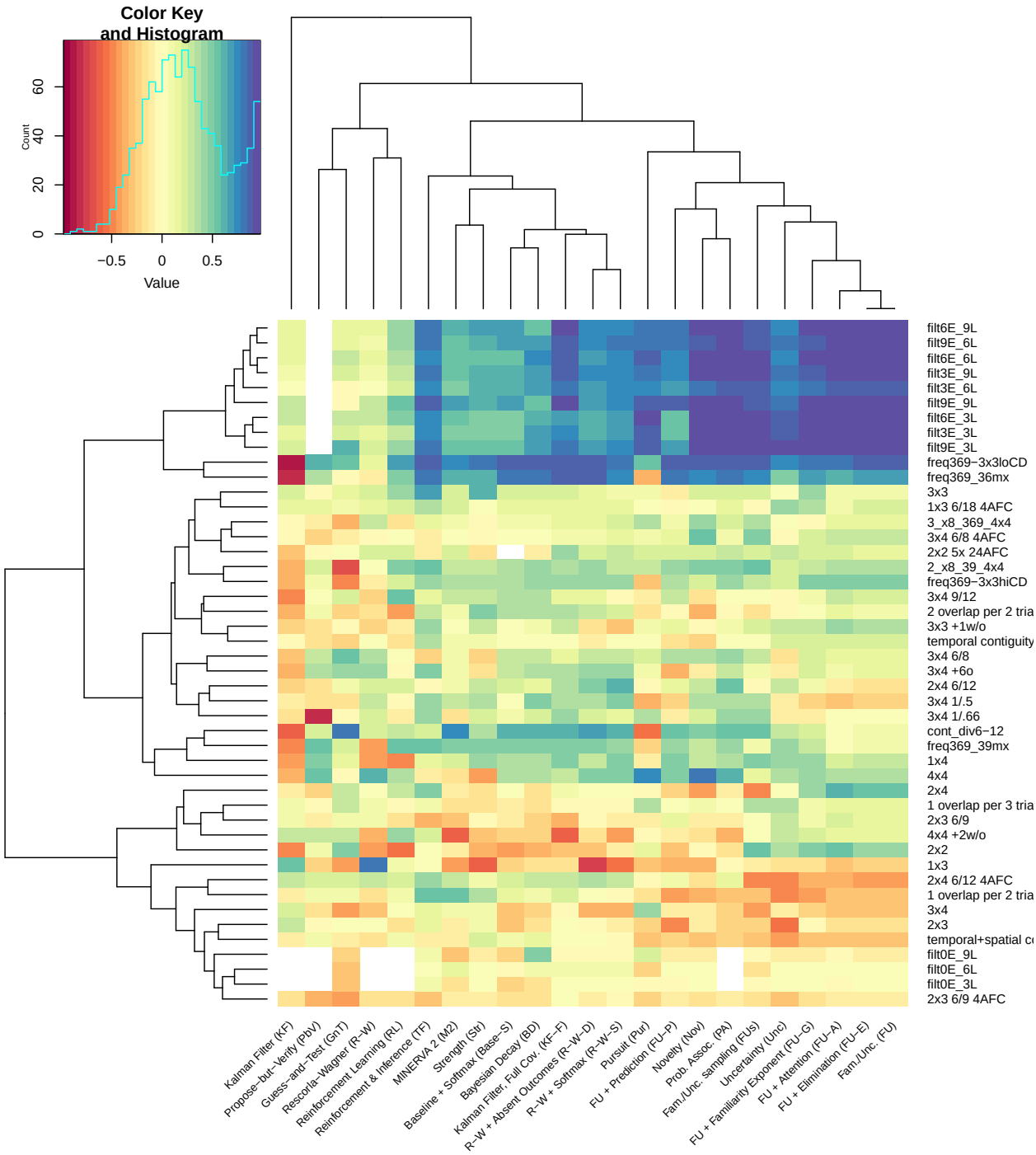
| Condition Name | Trials | Words/Trial | Objects/Trial | Items | Accuracy | SD | N |
|---|---|---|---|---|---|---|---|
| 3x4 | 36 | 3 | 4 | 18 | 0.19 | 0.12 | 25 |
| 3x4 1/.5 | 36 | 3 | 4 | 18 | 0.22 | 0.11 | 25 |
| 3x4 1/.66 | 36 | 3 | 4 | 18 | 0.21 | 0.09 | 25 |
| 3x4 +6o | 36 | 3 | 4 | 18 | 0.27 | 0.12 | 20 |
| 2x4 | 54 | 2 | 4 | 18 | 0.30 | 0.15 | 33 |
| 3x3 +1w/o | 54 | 3 | 3 | 18 | 0.17 | 0.08 | 39 |
| 4x4 +2w/o | 54 | 4 | 4 | 18 | 0.10 | 0.05 | 39 |
| 1x3 6/18 4AFC | 108 | 1 | 3 | 18 | 0.67 | 0.10 | 43 |
| 2x3 6/9 4AFC | 54 | 2 | 3 | 18 | 0.69 | 0.11 | 38 |
| 2x4 6/12 4AFC | 54 | 2 | 4 | 18 | 0.62 | 0.12 | 31 |
| 3x4 6/8 4AFC | 36 | 3 | 4 | 18 | 0.69 | 0.09 | 36 |
| 1x3 | 108 | 1 | 3 | 18 | 0.74 | 0.11 | 23 |
| 2x3 | 54 | 2 | 3 | 18 | 0.58 | 0.07 | 23 |
| 2x4 6/12 | 54 | 2 | 4 | 18 | 0.33 | 0.18 | 14 |
| 3x4 6/8 | 36 | 3 | 4 | 18 | 0.42 | 0.14 | 13 |
| 2x3 6/9 | 54 | 2 | 3 | 18 | 0.55 | 0.11 | 32 |
| 3x4 9/12 | 54 | 3 | 4 | 18 | 0.69 | 0.08 | 33 |
| 1x4 | 108 | 1 | 4 | 18 | 0.19 | 0.10 | 40 |
| 1x3 | 108 | 1 | 3 | 18 | 0.39 | 0.12 | 40 |
| 4x4 | 27 | 4 | 4 | 18 | 0.31 | 0.07 | 77 |
| 3x3 | 36 | 3 | 3 | 18 | 0.43 | 0.08 | 36 |
| 2x2 | 54 | 2 | 2 | 18 | 0.79 | 0.11 | 19 |
| 2x2 5x 24AFC | 60 | 2 | 2 | 24 | 0.51 | 0.12 | 46 |
| filt0E_3L | 18 | 2 | 2 | 24 | 0.37 | 0.11 | 31 |
| filt3E_3L | 27 | 2 | 2 | 24 | 0.45 | 0.29 | 30 |
| filt6E_3L | 36 | 2 | 2 | 24 | 0.42 | 0.30 | 30 |
| filt9E_3L | 45 | 2 | 2 | 24 | 0.41 | 0.30 | 31 |
| filt0E_6L | 36 | 2 | 2 | 24 | 0.49 | 0.10 | 23 |
| filt3E_6L | 45 | 2 | 2 | 24 | 0.48 | 0.22 | 23 |
| filt6E_6L | 54 | 2 | 2 | 24 | 0.55 | 0.27 | 23 |
| filt9E_6L | 63 | 2 | 2 | 24 | 0.51 | 0.24 | 23 |
| filt0E_9L | 54 | 2 | 2 | 24 | 0.59 | 0.11 | 31 |
| filt3E_9L | 63 | 2 | 2 | 24 | 0.58 | 0.27 | 31 |
| filt6E_9L | 72 | 2 | 2 | 24 | 0.59 | 0.26 | 31 |
| filt9E_9L | 81 | 2 | 2 | 24 | 0.60 | 0.27 | 31 |
| 2_x8_39_4x4 | 27 | 4 | 4 | 18 | 0.41 | 0.16 | 30 |
| 3_x8_369_4x4 | 27 | 4 | 4 | 18 | 0.39 | 0.13 | 30 |
| freq369-3x3loCD | 36 | 3 | 3 | 18 | 0.49 | 0.23 | 34 |
| freq369-3x3hiCD | 36 | 3 | 3 | 18 | 0.56 | 0.12 | 104 |
| freq369_36mx | 36 | 3 | 3 | 18 | 0.45 | 0.16 | 66 |
| freq369_39mx | 36 | 3 | 3 | 18 | 0.62 | 0.14 | 40 |
| cont_div6-12 | 36 | 3 | 3 | 18 | 0.43 | 0.11 | 62 |
| temporal+spatial contiguity | 27 | 4 | 4 | 18 | 0.49 | 0.08 | 30 |
| temporal contiguity only | 27 | 4 | 4 | 18 | 0.40 | 0.10 | 32 |
| 1 overlap per 3 trials, contr | 27 | 4 | 4 | 18 | 0.32 | 0.15 | 31 |
| 1 overlap per 2 trials, contr | 27 | 4 | 4 | 18 | 0.28 | 0.14 | 31 |
| 2 overlap per 2 trials, contr | 27 | 4 | 4 | 18 | 0.41 | 0.11 | 31 |

Table 1: Summary of modeled datasets.




## Correlation of model vs. human item-level performance per condition

Each cell displays the correlation coefficient of model vs. human item-level performance in a given experimental condition. 0 conditions (the Koehne et al. 2013 and Suanda et al. 2014 conditions, which only have condition-level human accuracy) are excluded here, since a correlation is undefined without item-level variance in the human data; they are included in the SSE-based heatmap above and in Table 2 below.




## Which experimental designs are easy or hard, overall?

The heatmaps above group misfit by model; here we instead average over models, to ask which experimental designs are uniformly easy or hard to predict for *any* model – a useful signal for choosing which conditions actually discriminate between theories, versus which conditions every model gets right (or wrong) regardless of mechanism.

Table 2 shows the five conditions with the lowest mean SSE across all 23 models (easiest, fit well by essentially every model) and the five with the highest (hardest, fit poorly by essentially every model). The `SD SSE`

column shows how much models disagree on a condition: a low SD alongside a high mean SSE means every model struggles with that design about equally – a design that is hard for principled reasons rather than one that merely discriminates between theories.

# Model disagreement: a discriminability index independent of human data

Mean SSE (above) indexes how *hard* a condition is, but hardness and discriminative *value* are different things: a condition every model gets equally wrong is hard but uninformative for choosing between theories, while a condition that splits the models – whoever turns out to be right – is exactly the kind of experiment worth running again with more power, or in a new population. We index this directly, independent of human data, as the standard deviation of each model's predicted item-level accuracy (`ModelPerf`, at each model's group-fit parameters) across models, averaged over the items in a condition. Because several of our associative models are one-parameter-removed cousins of each other (FU/FUs/FU-A/FU-P/Str/Unc/Nov all share the same core mechanism), their mutual spread can pad this index without reflecting genuine disagreement between *theories*. We therefore report it two ways: across all 24 fitted models, and across 12 representatives, one per distinct mechanism (Fam./Unc. (FU), Prob. Assoc. (PA), Bayesian Decay (BD), Rescorla-Wagner (R-W), Propose-but-Verify (PbV), Guess-and-Test (GnT), Pursuit (Pur), Reinforcement Learning (RL), Kalman Filter (KF), Reinforcement & Inference (TF), MINERVA 2 (M2), Baseline) – the two versions correlate at 0.99, but reorder some conditions meaningfully (see below).

Table 3 shows the most- and least-discriminative conditions by this measure. The `filt9E_*`/`filt6E_*` conditions – which present a group of pairs early, then systematically pair a subset of the late-stage objects with the early-stage words to test how strictly a mutual-exclusivity bias is maintained – are the most model-discriminating conditions in the whole set, on both versions of the index; the temporal- and spatial-contiguity conditions also rise near the top once the FU family's internal spread is removed, indicating a genuine cross-theory split there rather than one associative model disagreeing with its own close cousins. At the other end, `filt0E_6L`/`filt0E_9L` (the late-stage-only `filt` conditions, with no early pairs to violate) and several of the fixed-repetition `3x4`/`4x4` conditions are answered almost identically by every model regardless of mechanism – useful as sanity checks, but uninformative for adjudicating between theories. This index correlates with the SSE-based "hardest" ranking above at only $r \approx 0.75$, confirming that "hard for everyone" and "discriminates between theories" are related but distinct properties of a design.

## Each Model's Best- and Worst-Fitting Experiment

Table 4 summarizes for each model, what experimental condition is best fit by that model (in terms of correlation, `best` and `best_cond`), and what condition has the worst fit for that model (`worst` and `worst_cond`), again excluding the 0 conditions with no item-level human variance.

Across models, freq369-3x3loCD is most often the best-fit condition (best for 10 of 23 models), and 2x4 6/12 4AFC is most often the worst-fit condition (worst for 4 models).

## Best- and Worst-Fitting Models per Experiment

Table 5 takes the complementary view: for each experimental condition, which model fits it best and which fits it worst (by item-level correlation), and then, aggregating over conditions, how many conditions each model "wins" (is the single best-fitting model for) and "loses" (is the single worst-fitting model for). A model that wins many, varied conditions is a safe default; a model that wins a few conditions no other model wins is doing something distinctive worth understanding.

| Rank | Condition | Mean r | Mean SSE | SD SSE |
|---|---|---|---|---|
| Easiest | 4x4 +2w/o | -0.042 | 0.011 | 0.005 |
| Easiest | 3x3 | 0.219 | 0.011 | 0.006 |
| Easiest | 4x4 | 0.314 | 0.012 | 0.009 |
| Easiest | cont_div6-12 | 0.338 | 0.013 | 0.005 |
| Easiest | 3x4 +6o | 0.150 | 0.019 | 0.006 |
| Hardest | filt0E_3L | -0.043 | 0.132 | 0.169 |
| Hardest | filt0E_6L | 0.008 | 0.098 | 0.107 |
| Hardest | filt9E_3L | 0.725 | 0.096 | 0.127 |
| Hardest | filt3E_3L | 0.671 | 0.095 | 0.114 |
| Hardest | filt6E_3L | 0.702 | 0.093 | 0.121 |

Table 2: The five easiest and five hardest experimental conditions, by mean SSE across all fitted models.

| Rank | Condition | SD (all models) | SD (mechanism reps) |
|---|---|---|---|
| Most discriminative | filt9E_3L | 0.345 | 0.353 |
| Most discriminative | filt6E_3L | 0.326 | 0.337 |
| Most discriminative | filt9E_6L | 0.316 | 0.324 |
| Most discriminative | filt3E_3L | 0.304 | 0.318 |
| Most discriminative | filt6E_6L | 0.300 | 0.304 |
| Most discriminative | filt9E_9L | 0.300 | 0.304 |
| Least discriminative | 3x4 1/.66 | 0.068 | 0.072 |
| Least discriminative | 3x4 | 0.067 | 0.070 |
| Least discriminative | cont_div6-12 | 0.066 | 0.069 |
| Least discriminative | 4x4 +2w/o | 0.056 | 0.067 |

Table 3: Most- and least-discriminative experimental conditions: standard deviation of model-predicted item accuracy across models, at each model's group-fit parameters. 'all models' is 16 models (the 15-model roster plus baseline); 'mechanism reps' is one representative per distinct theoretical mechanism.

| Model | best | worst | best_cond | worst_cond |
|---|---|---|---|---|
| Fam./Unc. sampling (FUs) | 0.98 | -0.48 | filt6E_3L | 2x4 6/12 4AFC |
| FU + Familiarity Exponent (FU-G) | 0.97 | -0.42 | filt9E_9L | 1 overlap per 2 trials, contr |
| FU + Attention (FU-A) | 0.97 | -0.38 | filt6E_3L | 2x4 6/12 4AFC |
| Prob. Assoc. (PA) | 0.97 | -0.37 | filt9E_9L | 4x4 +2w/o |
| Fam./Unc. (FU) | 0.97 | -0.40 | filt9E_9L | 2x4 6/12 4AFC |
| FU + Elimination (FU-E) | 0.97 | -0.40 | filt9E_9L | 2x4 6/12 4AFC |
| Novelty (Nov) | 0.96 | -0.41 | filt9E_9L | 2x4 |
| Uncertainty (Unc) | 0.95 | -0.53 | filt9E_3L | 2x3 |
| Pursuit (Pur) | 0.95 | -0.53 | filt6E_3L | cont_div6-12 |
| Kalman Filter, Full Cov. (KF-F) | 0.93 | -0.63 | filt6E_9L | 4x4 +2w/o |
| Bayesian Decay (BD) | 0.90 | -0.36 | freq369-3x3loCD | 2x2 |
| Baseline + Softmax (Base-S) | 0.90 | -0.44 | freq369-3x3loCD | 2x2 |
| R-W + Absent Outcomes (R-W-D) | 0.89 | -0.74 | freq369-3x3loCD | 1x3 |
| FU + Prediction (FU-P) | 0.87 | -0.48 | freq369-3x3loCD | 2x3 |
| Reinforcement & Inference (TF) | 0.85 | -0.39 | freq369-3x3loCD | 2x3 6/9 |
| Strength (Str) | 0.84 | -0.59 | freq369-3x3loCD | 1x3 |
| Rescorla-Wagner (R-W) | 0.84 | -0.41 | 1x3 | 1x4 |
| R-W + Softmax (R-W-S) | 0.82 | -0.57 | freq369-3x3loCD | 1x3 |
| Guess-and-Test (GnT) | 0.79 | -0.69 | cont_div6-12 | 2_x8_39_4x4 |
| MINERVA 2 (M2) | 0.77 | -0.62 | freq369-3x3loCD | 4x4 +2w/o |
| Reinforcement Learning (RL) | 0.71 | -0.54 | freq369-3x3loCD | 2x2 |
| Propose-but-Verify (PbV) | 0.60 | -0.83 | freq369-3x3loCD | 3x4 1/.66 |
| Kalman Filter (KF) | 0.58 | -0.90 | 1x3 | freq369-3x3loCD |

Table 4: Each model's best- and worst-fitting experiment.

| Model | best_fits | worst_fits |
|---|---|---|
| Reinforcement & Inference (TF) | 7 | 1 |
| FU + Attention (FU-A) | 5 | 0 |
| Kalman Filter, Full Cov. (KF-F) | 4 | 1 |
| Bayesian Decay (BD) | 3 | 0 |
| FU + Elimination (FU-E) | 3 | 0 |
| Fam./Unc. sampling (FUs) | 3 | 2 |
| Novelty (Nov) | 3 | 0 |
| Reinforcement Learning (RL) | 3 | 3 |
| FU + Familiarity Exponent (FU-G) | 2 | 0 |
| Guess-and-Test (GnT) | 2 | 10 |
| Pursuit (Pur) | 2 | 1 |
| Strength (Str) | 2 | 1 |
| Uncertainty (Unc) | 2 | 4 |
| FU + Prediction (FU-P) | 1 | 2 |
| Kalman Filter (KF) | 1 | 10 |
| MINERVA 2 (M2) | 1 | 2 |
| R-W + Softmax (R-W-S) | 1 | 1 |
| Rescorla-Wagner (R-W) | 1 | 5 |
| Propose-but-Verify (PbV) | 0 | 2 |
| R-W + Absent Outcomes (R-W-D) | 0 | 1 |

Table 5: Number of experimental conditions that each model fits best, and worst (by item-level correlation).

# Appendix C: Model and Parameter Recovery

Two questions bear on how far the comparison can be trusted. First, if one of these models had generated the data, would fitting all of them have identified it, or do some mechanisms mimic each other so closely on these 47 designs that the comparison could not tell them apart? Second, are a model's fitted parameters interpretable, or could quite different parameter values produce the same predictions? We answered both with simulation, using a reduced version of the fitting procedure in the main text.

**Synthetic data.** A generating model was run on all 47 conditions (with 500 simulated learners for stochastic models, to approximate its true predictions closely); each item's synthetic accuracy was then drawn as a binomial proportion with that item's real number of participants. Synthetic data are therefore exactly as noisy as the human data under the binomial noise model behind the noise ceiling, though they lack participant-level individual differences.

**Fitting.** Synthetic data were fit with the same objective (participant-weighted SSE) and parameter bounds as the main analysis, but a reduced differential-evolution budget, so that several hundred fits were affordable: a population of 10 per free parameter (at least 20), and 60 generations for deterministic models or 25 for stochastic ones, whose fits used 50 simulated learners per evaluation. We used one representative of each mechanism: PA, FU, FUs, BD, TF, M2, RL, and PbV, plus the baseline in model recovery.

**Parameter recovery.** For each of the 8 models, 20 parameter vectors were drawn uniformly within the fitting bounds (log-uniformly for scale parameters whose bounds span two or more orders of magnitude), discarding draws whose mean predicted accuracy fell below .1 or above .95, since such designs carry little information. Each synthetic dataset was then fit by the model that generated it.

**Model recovery.** Each of the 9 models generated 5 synthetic datasets at its group-fit parameters from the main analysis, and every dataset was fit by all 9 models. The winner was the model with the lowest objective; stochastic models were re-evaluated with 200 simulated learners at their best parameters, since the minimum the optimizer reports for a noisy objective is biased low.

## Results

**Parameters are mostly recoverable.** Figure 1 plots recovered against generating values, and Table 6 summarizes them. 18 of the 19 parameters were recovered with $r \geq .70$ (on a log scale for the log-scaled ones), and 9 with $r \geq .95$. No recovered value fell on a fitting bound. There are three caveats.

- **The propose-but-verify model's recall probability $\alpha$ is not identifiable** ($r = .33$): recovered values cluster near .5 whatever the generating value, plausibly because $\alpha$ and the recall increment $\epsilon$ both control how often a proposed referent survives to be verified, so different combinations predict similar accuracy. PbV's fitted $\alpha$ should not be interpreted on its own.
- **PA's smoothing constant $\lambda$ is identifiable only when it is not negligible.** Above about $10^{-4}$ it is recovered closely (median error 0.03 $\log_{10}$ units); below that, any small value predicts the same accuracies, so the recovered value is arbitrary.
- **Several parameters are recovered only loosely:** BD's $\tau$ (.74), FUs' $\lambda$ (.78), RL's $\beta$ (.80), BD's $\delta$ (.80), PbV's $\epsilon$ (.82), and BD's $\alpha$ (.83). Their occasional large misses (Figure 1) suggest partial trade-offs with the same model's other parameters, so these values should be interpreted with caution.

**Models are perfectly recoverable at their fitted parameters.** In every one of the 45 synthetic datasets, the generating model fit best (45 of 45; Figure 2). For every generator except the baseline, the closest rival's objective exceeded the generator's by at least .07 (Figure 3). The baseline is the hardest case: its data were best fit by the baseline every time, but by margins of only .01–.02 over the reinforcement-learning model. The closest rivals are mostly the expected ones: FU and FUs for each other; PA for BD, TF and M2; M2 for RL; and RL for PbV, the two models that commit to one referent per word on each trial.

**What this does and does not show.** Perfect recovery means that the 47 designs, at their real sample sizes, are informative enough to distinguish these mechanisms *when the data sit where each model's best fit to the human data puts them.* It does not show that every region of each model's parameter space is distinguishable, and the reduced fitting budget and in-sample selection make it, if anything, a lenient test.

Nor does it make the human-data ranking any less dependent on the models' specific assumptions: a model can be recoverable and still be wrong.

Table 6: Parameter recovery: correlation between generating and recovered values over 20 synthetic datasets per model, and the median absolute error, both on the scale shown (natural log for log-scaled parameters).

| Model | Parameter | Scale | r | Median abs. error |
|---|---|---|---|---|
| PA | $\lambda$ | log | .90 | 0.69 |
| PA | $\beta$ | log | .99 | 0.099 |
| FU | $\chi$ | log | 1.00 | 0.026 |
| FU | $\lambda$ | log | 1.00 | 0.039 |
| FU | $\alpha$ | linear | 1.00 | 0.0011 |
| FUs | $\chi$ | log | .99 | 0.17 |
| FUs | $\lambda$ | log | .78 | 0.19 |
| FUs | $\alpha$ | linear | .99 | 0.0073 |
| BD | $\alpha$ | log | .83 | 0.065 |
| BD | $\delta$ | log | .80 | 0.12 |
| BD | $\tau$ | log | .74 | 0.14 |
| TF | $x$ | linear | .96 | 0.014 |
| TF | $b$ | linear | .93 | 0.014 |
| TF | $\alpha_0$ | linear | .99 | 0.006 |
| M2 | $X$ | log | .98 | 0.064 |
| RL | $\eta$ | linear | .88 | 0.058 |
| RL | $\beta$ | log | .80 | 0.35 |
| PbV | $\alpha$ | linear | .33 | 0.18 |
| PbV | $\epsilon$ | linear | .82 | 0.15 |

## Error-driven models

The Rescorla-Wagner model was refit after a bug was corrected in its update (in the version first fit, each word received the prediction error of a different object, which reduced the model to co-occurrence counting). We also added three error-driven models that follow the original formulations more closely, all with a softmax choice rule at test ($P(o|w) \propto \exp(\phi\, M_{w,o})$, inverse temperature $\phi$) in place of Luce choice: R-W-S, the Rescorla-Wagner model with a softmax choice rule; R-W-D, which also treats the objects absent from a trial as outcomes with target 0, as in Rescorla and Wagner (1972), so that associations with absent objects are unlearned; and KF-F, a Kalman filter with full covariance among the words predicting each object and absent objects as outcomes (Dayan & Kakade, 2001), in which a word's association with an object can change on trials where that word is not heard (e.g., backward blocking). In all three, the salience and the asymptote are fixed at 1 and the observation noise at 1, since they only rescale the other parameters; the free parameters are the learning rate $\beta$ (or, for KF-F, the process noise $\tau^2$ and initial uncertainty $\sigma_0^2$), the decay $\alpha$ (R-W-S and R-W-D), and $\phi$. Base-S, the co-occurrence baseline with the same softmax rule, is the control: it shows how much of the variants' improvement comes from the choice rule alone.

These models were recovered in a separate run with the same procedure (Table 7; Figures 4–5): 20 parameter draws per model, and model recovery among the 8 models Base, PA, FU, Base-S, R-W, R-W-S, R-W-D, and KF-F, with 5 synthetic datasets per generator. All are deterministic, so every fit used the deterministic budget.

**The softmax models are identifiable.** Every parameter of Base-S, R-W-S, R-W-D, and KF-F was recovered with $r \geq .95$, except KF-F's process noise $\tau^2$ ($r = .80$), whose two large misses came from draws with very small initial uncertainty $\sigma_0^2$, where diffusion has little effect on the predictions. In model recovery,

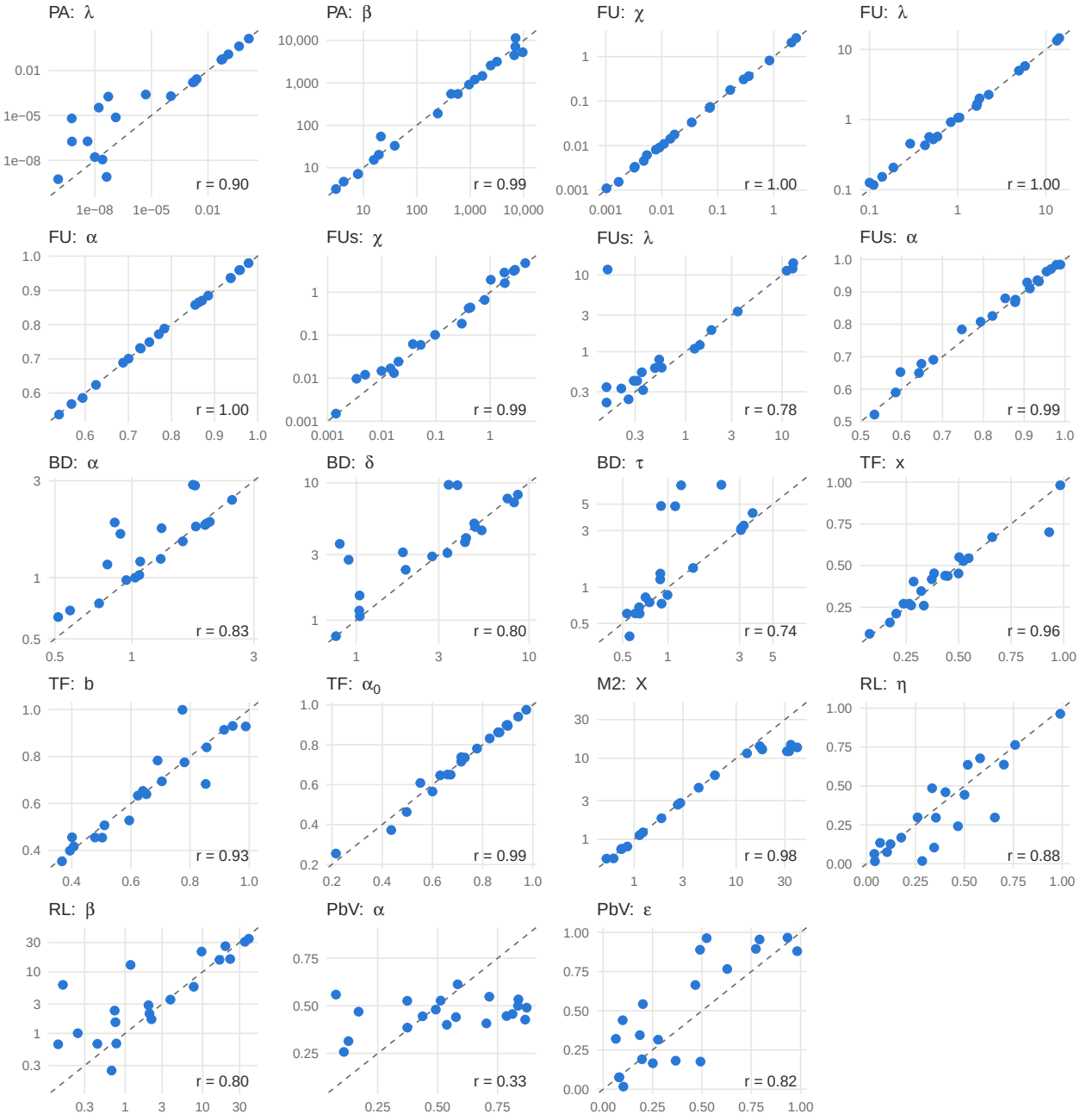



Figure 1: Parameter recovery. Each panel is one parameter of one model; each point is one synthetic dataset, generated at the value on the x axis and fit by the same model. Points on the dashed identity line are recovered exactly.

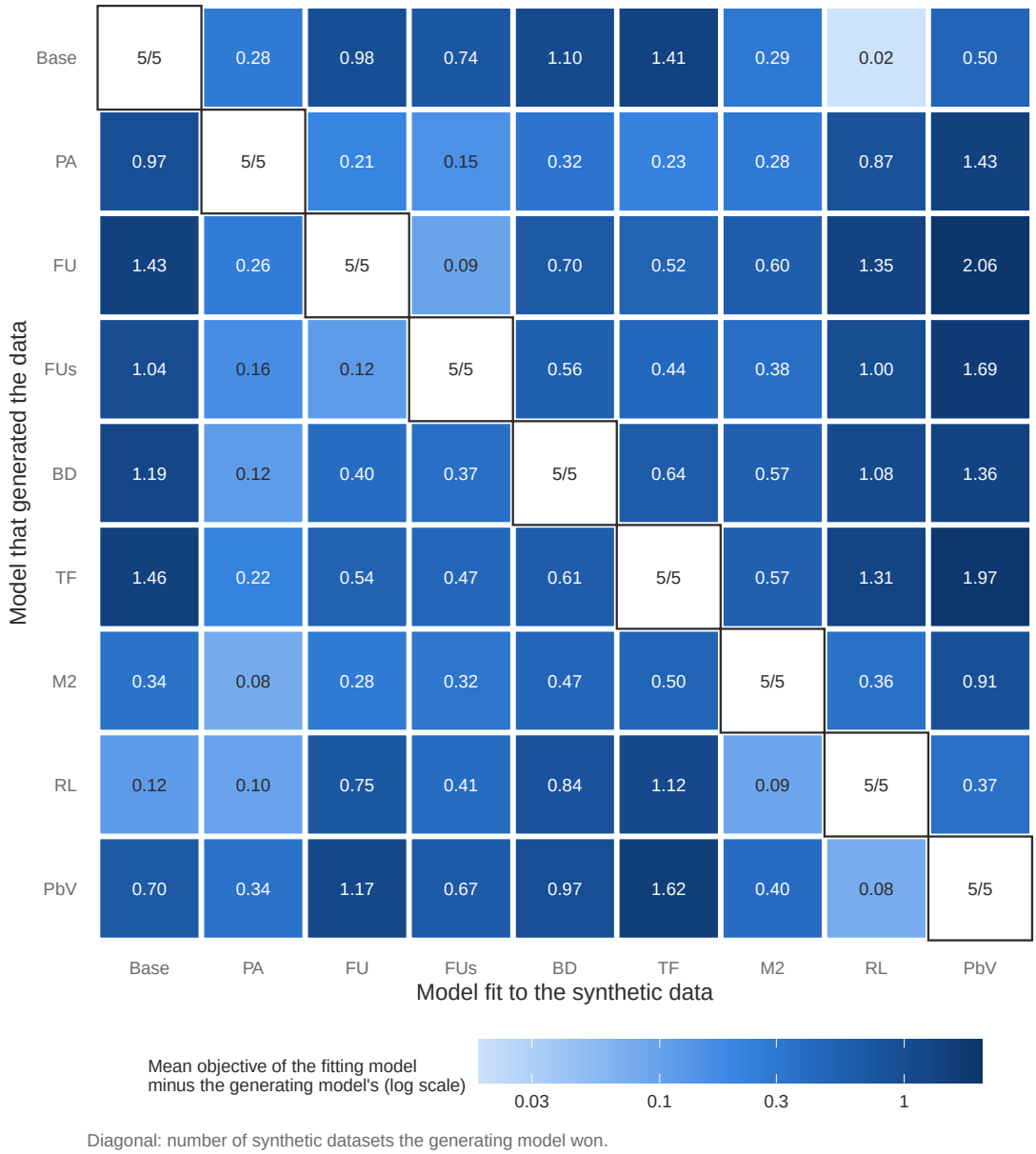



Figure 2: Model recovery. Rows: the model that generated the synthetic data (at its group-fit parameters); columns: the model fit to them. Off-diagonal cells give the mean amount by which the fitting model's objective exceeded the generating model's (darker = easier to tell apart); the diagonal gives the number of datasets the generating model won.

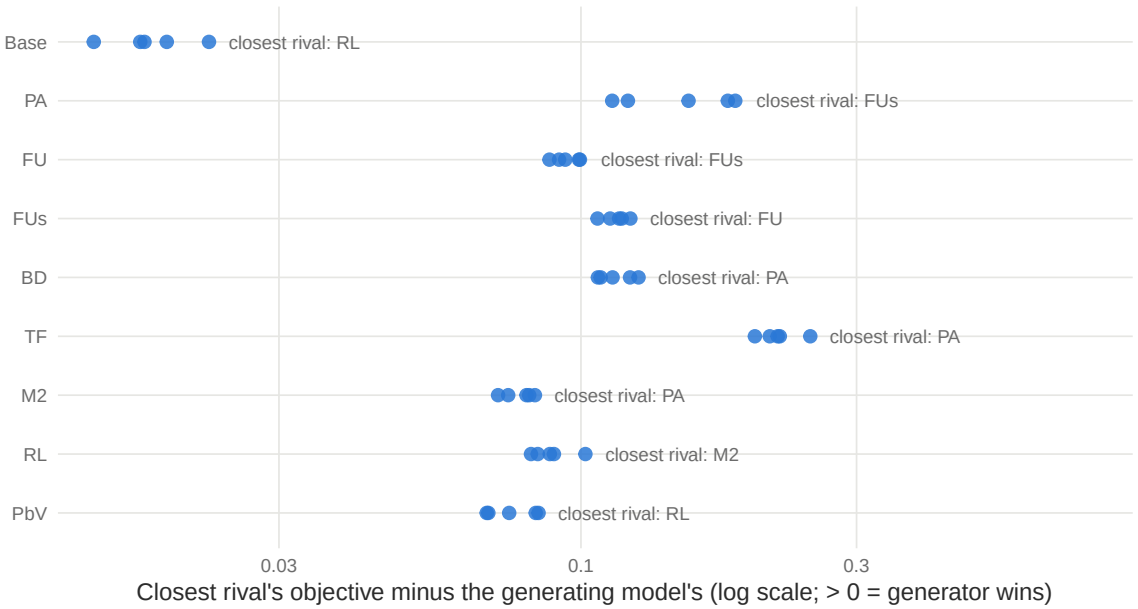
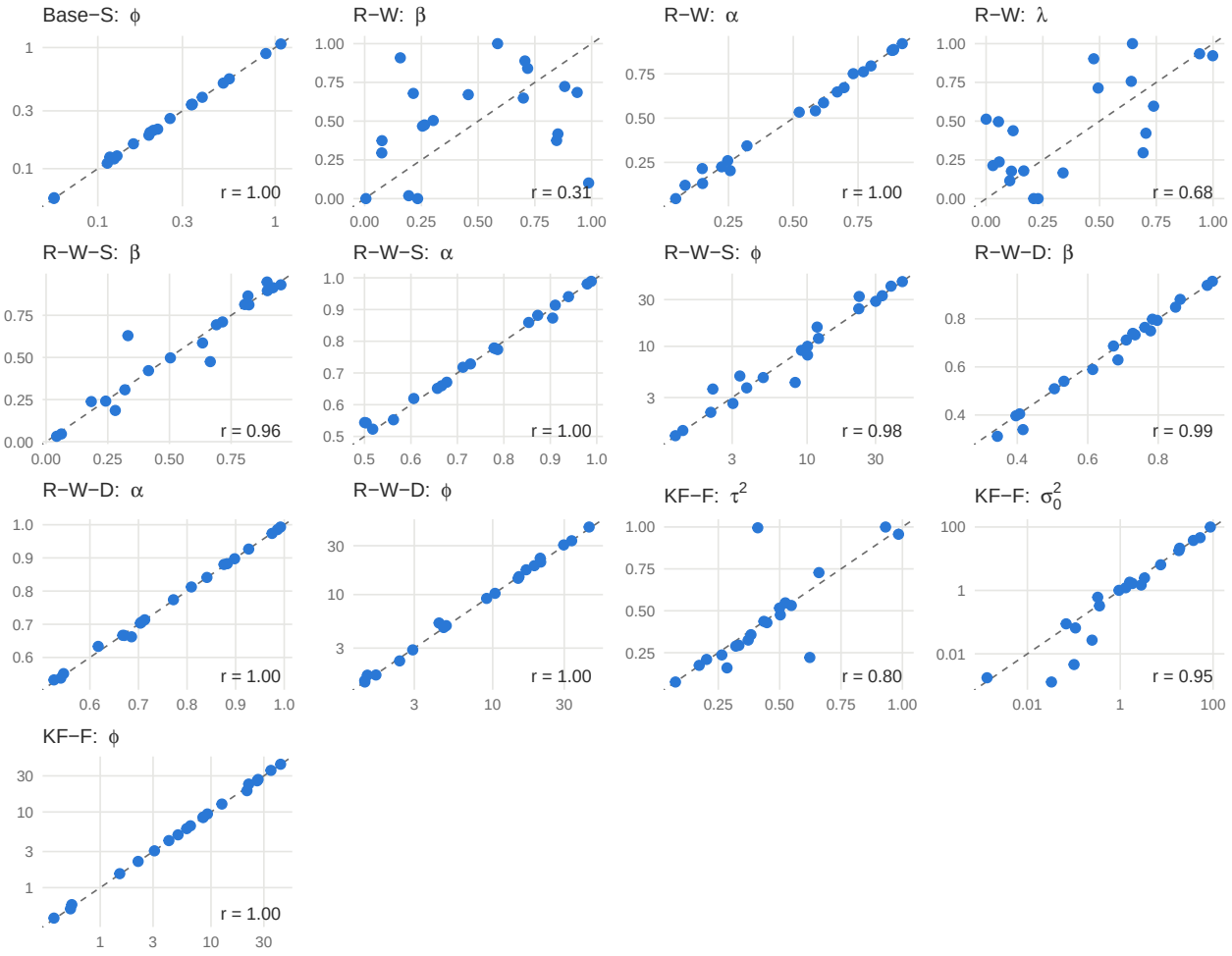



Figure 3: Margin of victory: for each synthetic dataset, how much worse the best-fitting rival model's objective was than the generating model's, with the rival that came closest most often.

each of them won all 5 of its own datasets, by margins of .01–.05 over its closest rival, usually another error-driven softmax model. These margins are smaller than those separating the main models (at least .07 above), so the 47 designs distinguish the error-driven variants from one another, but only just.

**The corrected Rescorla-Wagner model with Luce choice is not.** Its decay $\alpha$ is recovered ($r = 1.00$), but its learning rate $\beta$ ($r = .31$) and asymptote $\lambda$ ($r = .68$) are not, nor is their product, the effective learning rate ($r = .37$ on a log scale); 15% of fits put one of them on a fitting bound. In model recovery, R-W and the baseline cannot be told apart: on each other's data their objectives differed by at most .0009, and the baseline won only 1 of its own 5 datasets, the rest going to R-W's extra free parameters. Under Luce choice, then, the learning rate barely changes this model's predictions on these designs, and its fit should be read as the baseline's. The error-driven mechanism is distinguishable only with a choice rule that is sensitive to the scale of the learned associations.




x: generating (true) value; y: recovered value; dashed: identity. Log axes where the fitting bounds span 100x or more.

Figure 4: Parameter recovery for the error-driven models. As in Figure 1.

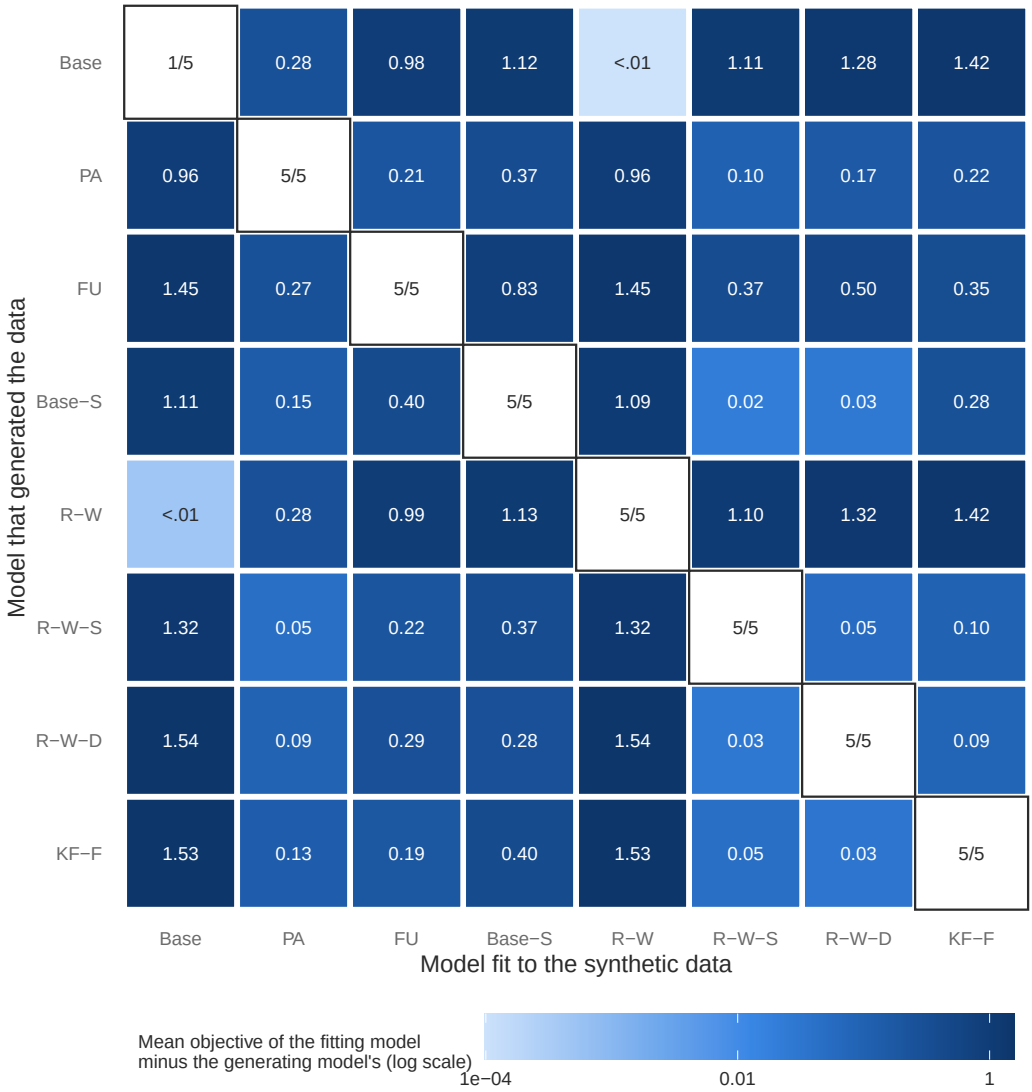



Figure 5: Model recovery for the error-driven models. As in Figure 2. Base and R-W are indistinguishable (gaps under .01 in both directions); every other generator won all its datasets.

Table 7: Parameter recovery for the error-driven models, as in Table 6: 20 synthetic datasets per model.

| Model | Parameter | Scale | r | Median abs. error |
|---|---|---|---|---|
| Base-S | $\phi$ | log | 1.00 | 0.0073 |
| R-W | $\beta$ | linear | .31 | 0.22 |
| R-W | $\alpha$ | linear | 1.00 | 0.016 |
| R-W | $\lambda$ | linear | .68 | 0.2 |
| R-W-S | $\beta$ | linear | .96 | 0.011 |
| R-W-S | $\alpha$ | linear | 1.00 | 0.0059 |
| R-W-S | $\phi$ | log | .98 | 0.036 |
| R-W-D | $\beta$ | linear | .99 | 0.0059 |
| R-W-D | $\alpha$ | linear | 1.00 | 0.001 |
| R-W-D | $\phi$ | log | 1.00 | 0.016 |
| KF-F | $\tau^2$ | linear | .80 | 0.027 |
| KF-F | $\sigma_0^2$ | log | .95 | 0.14 |
| KF-F | $\phi$ | log | 1.00 | 0.01 |